Computer Networks

Unit-wise Previous Year Questions · CBIT (Autonomous) · R18 / R20 / R22 · 2021–2025

Unit	Questions	Most repeated topics (focus here first)
UNIT I Introduction & Physical Layer	60	OSI model, TCP/IP model & comparison, circuit vs packet switching, topologies + mesh cable numericals, guided media / fiber optics
UNIT II Data Link Layer & Medium Access Sub-layer	52	CRC numericals (10011101 / x^3+1 repeated), Hamming code & distance, GBN vs SR + window numericals, CSMA variants & CSMA/CD, collision-free protocols
UNIT III Network Layer	60	Distance vector & count-to-infinity, link state, DV vs LS, Dijkstra, flooding, NAT, subnetting numericals, IPv6 vs IPv4, leaky/token bucket
UNIT IV Transport Layer	58	TCP segment header, 3-way handshake / connection establishment & release, TCP vs UDP + header comparison, UDP header, multiplexing, congestion control
UNIT V Application Layer	36	DNS (components, resource records, iterative vs recursive), e-mail architecture (UA/MTA), SMTP / POP3 / IMAP, FTP, SNMP

Paper codes used in this booklet

Tag	Exam	Code	Reg.	Branch / Sem
N25	Nov 2025	22ITC10	R22	CSE/IT/AI&DS - V Sem
M25	May 2025	22CAC06	R22	CSM/AI&ML - VI Sem (DCCN)
D24	Dec 2024	22ITC10	R22	CSE/IT/AI&DS - V Sem
J24	May-Jun 2024	20CSC21	R20	CSM/AI&ML - VI Sem (DCCN)
D23-IT	Dec 2023	20ITC20	R20	IT/AI&DS - V Sem
D23-CET	Dec 2023	20CIC04	R20	CET - V Sem
D23-CSE	Dec 2023	20CSC21	R20	CSE - V Sem (DCCN)
J23	May-Jun 2023	20CSC21	R20	CSE-AI&ML - VI Sem (DCCN)
D22-IT	Dec 2022	20ITC20	R20	IT/AI&DS - V Sem
D22-IoT	Dec 2022	20CIC04	R20	CSE-IoT&CS/BCT - V Sem
D22-CSE	Dec 2022	20CSC21	R20	CSE - V Sem (DCCN)
D22-ECE	Dec 2022	18ECC26	R18	ECE - VII Sem
J22	Jun 2022	18CSC23	R18	CSE - VI Sem (DCCN)
D21-ECE	Dec 2021	18ECC26	R18	ECE - VII Sem
D21-IT	Dec 2021	18ITC17	R18	IT - V Sem

How to read: M = marks. Same/near-identical questions from different papers are merged into one row, with all papers listed. **[HOT]** marks topics asked 4+ times. **[Numerical]** marks problems to practise. Questions on topics outside your syllabus (WWW, HTTP, streaming, security, HDLC, etc.) are collected in the Appendix.

UNIT I · Introduction & Physical Layer

Network Basics, Hardware & Applications *[HOT] asked 12 times*

#	Question	M	Paper(s)
1	What is a Computer Network? Explain the applications of Computer Networks.	4	N25
2	Define the following terms: (i) Computer network (ii) Protocol.	2	J22
3	Explain briefly about the Internet.	2	D21-ECE
4	Distinguish between host and client.	2	D22-ECE, D21-ECE
5	What is the purpose of an NIC?	2	M25
6	Explain LAN, MAN and WAN technologies with suitable architecture diagrams.	5	D23-CET
7	An alternative to a LAN is a big timesharing system with terminals for all users. Give two advantages of a client-server system using a LAN.	5	D22-IoT
8	What are the advantages of a multipoint connection over a point-to-point connection?	3	J24
9	When a party makes a local telephone call to another party, is this a point-to-point or multipoint connection? Explain.	5	D22-CSE
10	Name any two reasons for using layered protocols.	3	D22-IoT
11	With a neat diagram describe the Data Communication Model.	5	J22

Network Topologies *[HOT] asked 9 times*

#	Question	M	Paper(s)
12	Define bus, ring, star and mesh topologies. Provide one example of where each might be used.	5	D24
13	List out the four basic network topologies.	2	J22
14	Name the four basic network topologies, and cite an advantage of each type.	3	D22-CSE
15	Describe bus topology with a suitable example and give its advantages and disadvantages.	4	D23-CSE
16	**[Numerical]** 7 devices are arranged in a mesh topology. How many cables are needed and how many ports for each device?	2	N25
17	**[Numerical]** 6 devices are arranged in a mesh topology. How many cables and ports per device?	3	J23
18	**[Numerical]** Number of cable links required for mesh topology? If 5 nodes are connected in a mesh, how many cables?	3	D23-CSE
19	**[Numerical]** For n devices in a network, what is the number of cable links required for mesh, ring, bus and star topology?	5	M25
20	**[Numerical]** Three packet-switching networks of n nodes: star (central switch), bidirectional ring, fully interconnected. Best-, average- and worst-case paths in hops?	5	J23

OSI Reference Model *[HOT] asked 7 times*

#	Question	M	Paper(s)
21	Draw and explain in detail the ISO-OSI Reference Model.	6	N25
22	Draw the complete OSI reference model layered architecture and discuss the responsibilities of each layer.	9	D23-CSE
23	What are the different layers in the OSI Reference Model? Explain the functions of each layer.	5	D22-IT
24	Illustrate the ISO-OSI Reference model. Explain the critique of the OSI Model and Protocols.	4	D22-CSE
25	Identify the functions of the physical and data link layers in the OSI model.	5	D24
26	Explain the functionalities of session layer and presentation layer in OSI reference model.	4	D23-CET
27	Explain the functionalities of network layer and transport layer in OSI reference model.	4	D23-CET

TCP/IP Reference Model *[HOT] asked 7 times*

#	Question	M	Paper(s)
28	Analyze the four layers of the TCP/IP reference model and explain the role of each layer in communication across networks.	5	D24
29	Discuss the TCP/IP reference model with a neat sketch in detail.	5	D23-IT
30	Explain the TCP/IP reference model with a neat sketch.	5	J24
31	Illustrate the layer hierarchy of the TCP/IP model with a neat sketch; explain each layer; compare OSI and TCP/IP.	5	J23
32	What are the five layers in the Internet protocol stack? What are the principal responsibilities of each?	5	D21-IT
33	Discuss / explain the services provided by the internet layer(s).	5	D22-ECE, D21-ECE

Comparison of OSI and TCP/IP *[HOT] asked 5 times*

#	Question	M	Paper(s)
34	Compare OSI and TCP/IP Reference Models.	3	D22-IT
35	Explain the differences between the TCP/IP and the OSI model.	5	M25
36	List two ways in which OSI and TCP/IP models are the same, and two ways in which they differ.	4	D22-IoT
37	Compare the performance of TCP/IP and ISO/OSI reference model.	5 / 4	J22, J24

Packet, Circuit & Virtual Circuit Switching *[HOT] asked 17 times*

#	Question	M	Paper(s)
38	Illustrate packet switching, circuit switching and virtual circuit switching with suitable diagrams.	6	N25
39	Explain circuit switching and packet switching.	5	D23-CET
40	What is packet switching? Compare and contrast a circuit-switched network to a packet-switched network.	5	D22-IoT
41	Compare and contrast a circuit-switched network and a packet-switched network.	5	M25, D21-IT
42	Compare circuit- and packet-switched networks. What is the role of the address field in a packet travelling through a virtual-circuit network?	4	D22-CSE
43	Explain the different types of switching networks; mention their advantages and disadvantages.	5+4	J24
44	Illustrate a simple circuit switching network with a neat diagram.	5	D22-ECE, D21-ECE
45	Illustrate a simple packet switching network with a neat diagram.	5	D22-ECE, D21-ECE

#	Question	M	Paper(s)
46	Write any four important points about circuit switching.	2	D22-ECE
47	Distinguish between connectionless and connection-oriented services of the Internet.	5	D22-ECE, D21-ECE
48	Describe circuit switched networks, datagram networks and cable networks with suitable diagrams.	5	J22
49	**[Numerical]** One store-and-forward packet switch between hosts, link rates R1 and R2. Total end-to-end delay for a packet of length L?	2	D21-IT
50	**[Numerical]** 1000-byte packet, link 2500 km, propagation speed 2.5×10^8 m/s, rate 2 Mbps: propagation delay? General formula? Does it depend on L or R?	5	D21-IT

Physical Layer: Transmission Media *[HOT] asked 10 times*

#	Question	M	Paper(s)
51	How does a twisted pair cable reduce interference?	2	D24
52	How do guided media differ from unguided media?	2	M25
53	Discuss the advantages of fiber optic cables. Explain with a neat sketch.	5	D24
54	Draw a side view of a fiber optic cable and explain the components.	3	D23-CET
55	Explain the two major categories of transmission media. Write the advantages of optical fiber over twisted-pair and coaxial cable.	5	D22-CSE
56	Clearly explain all guided media used for data (analog/digital) transmission.	4	D22-IoT
57	Discuss guided media with real-time applications: twisted pair, coaxial cable, fiber optic cable.	5	D23-CSE
58	Explain the three types of guided media and two types of unguided media.	5	J22
59	Discuss in detail un-guided transmission media.	4	N25
60	How does sky propagation differ from line-of-sight propagation? Omnidirectional vs unidirectional waves?	4	J23

UNIT II · Data Link Layer & Medium Access Sub-layer

Design Issues, Services & Framing *[HOT] asked 10 times*

#	Question	M	Paper(s)
1	What are the design issues of the Data Link Layer?	5	J22
2	What are the services provided by the Data Link Layer?	2	D22-ECE
3	List and explain the functionalities of a Data Link Layer.	3	D23-CET
4	Discuss framing in the data link layer.	2	N25
5	How would you apply framing in the DLL to differentiate between types of data, and what techniques implement it effectively?	5	D24
6	Design two simple algorithms for byte-stuffing: one adds bytes at the sender, the other removes them at the receiver.	4	J24
7	Compare and contrast flow control and error control.	5	D22-IoT
8	Differentiate flow control and error control. Discuss the importance of error control in transmission.	3	D23-CSE
9	If the data link layer can detect errors between hops, do we need another checking mechanism at the transport layer? Justify.	5	M25
10	Explain briefly why both MAC and IP addressing schemes are required in the same network.	3	D23-IT

Error Detection & Correction (Theory) *[HOT] asked 10 times*

#	Question	M	Paper(s)
11	Why is error detection important in the Data Link Layer?	2	D24
12	What is the need of Error Detection and Correction in the Data Link Layer?	2	D22-ECE
13	Explain single-bit errors and burst errors with examples.	3	D23-CET, J24
14	What is the Hamming distance? What is the minimum Hamming distance?	3 / 5	D22-IoT, J24
15	Outline the use of Cyclic Redundancy Check.	3	D23-IT
16	Explain how CRC detects errors in transmitted data, with an example.	5	D24
17	Analyze the CRC technique for error detection and correction with an example.	7	D23-CET
18	Explain checksum (error detection code) with an example.	5	J23

Error Detection & Correction (Numericals) *[HOT] asked 10 times*

#	Question	M	Paper(s)
19	CRC: bit stream 10011101, generator x^3+1. Find the transmitted string. If the 3rd bit from left is inverted, show how the receiver detects it.	5	N25, D22-CSE
20	CRC: message 11001001, polynomial x^3+1. What message should be transmitted?	4	D23-IT
21	CRC: 8-bit message 01010010, generator x^3+x+1. How is the message transmitted?	2	D23-CET
22	CRC: dataword 1010011110, divisor 10111. Generate the codeword (binary division); show checking at receiver. Relationship between degree of generator and divisor size.	4 / 5	D22-IoT, J23
23	CRC: frame 100110011011, $G(x) = x^4+x+1$. Find the encoded word and check for errors at receiver.	5	D21-IT
24	Hamming code: 16-bit messages. How many check bits for single-bit correction? Show the bit pattern for 1101110000110101 (odd parity).	5	D23-IT
25	Hamming code: 8-bit data frame. Find padding (check) bits, generate the frame; 4th bit inverted - find and correct the error.	6	D23-CSE
26	Find Hamming distance: d(10000,00000), d(10101,10000), d(11111,11111), d(000,000).	4	J24

Elementary & Sliding Window Protocols (Stop-and-Wait, GBN, SR) *[HOT] asked 12 times*

#	Question	M	Paper(s)
27	Demonstrate the functioning of a one-bit sliding window protocol with an example.	5	N25
28	Discuss sliding window protocols.	7	D23-CET
29	Explain the operation of the Go-Back-N protocol in reliable data transmission.	5	D24
30	Compare Go-Back-N ARQ protocol to Selective-Repeat ARQ.	5	D22-IoT
31	What is the advantage of Selective Repeat ARQ over Go-Back-N ARQ?	3	D23-CSE
32	Compare Stop-and-Wait ARQ with sliding window ARQ.	5	J22
33	What are the three protocols used for noisy channels? Explain.	5	J24
34	How are flow control services provided by the DLL? Compare with Protocol 5 (Go-Back-N) and Protocol 6 (Selective Repeat).	4	J23
35	In rdt protocols, why do we need sequence numbers and timers?	2	D21-IT
36	**[Numerical]** Go-Back-N, window 7, 1000 bits/packet, send 1 million bits, distance 5000 km, speed $2x10^8$ m/s. How long does it take?	4	D22-CSE
37	**[Numerical]** 15 packets, window 5, every 4th packet lost (no ACKs lost). Total transmissions with Go-Back-N?	5	M25
38	**[Numerical]** Go-Back-4 and Selective Repeat (Ws=3): every 6th packet lost, 10 packets to send. Total transmissions?	5	D21-IT

Channel Allocation, ALOHA & Random Access *[HOT] asked 6 times*

#	Question	M	Paper(s)
39	What are the various multiple access channels?	2	D21-ECE
40	Mention the advantages of slotted ALOHA protocol.	2	D21-ECE
41	Explain in detail / describe random access (control) protocols.	5	D22-ECE, D21-ECE
42	Explain channel partitioning MAC protocols.	5	D22-ECE, D21-ECE

CSMA, CSMA/CD, CSMA/CA & Collision-free Protocols *[HOT] asked 12 times*

#	Question	M	Paper(s)
43	Explain the concept of CSMA and its variants.	5	D24
44	Explain the three persistence methods of CSMA.	5	D22-IT
45	Discuss Carrier Sense Multiple Access protocols in detail.	4	D23-IT
46	What is CSMA/CD? How does it work? / Explain CSMA/CD in detail.	5	D23-CSE, D21-IT
47	Compare and contrast CSMA/CD with CSMA/CA.	5	M25
48	What are collision-free protocols? Explain them (with examples).	4	D22-IoT, D22-IT
49	Differentiate between collision-free protocols and collision-detection-based protocols with examples.	5	N25
50	**[Numerical]** Bandwidth $20x10^6$ bps, CSMA/CD, max propagation time 40 microseconds. Find minimum frame size (bytes).	2	D23-CET
51	**[Numerical]** 10 Mbps Ethernet, 2.5 km coax, speed $2.3x10^8$ m/s, 128-byte packet with 30 bytes overhead. Effective transmission rate and max throughput?	4	D22-IT
52	**[Numerical]** 10 Mbps Ethernet, medium 2500 m, frame 512 bytes, speed $2x10^8$ m/s. Find the transmission delay.	5	M25

UNIT III · Network Layer

Network Layer Design Issues *[HOT] asked 6 times*

#	Question	M	Paper(s)
1	Discuss the design issues in the Network Layer of the OSI model.	5	D24
2	Analyze how the Network Layer interacts with the Data Link and Transport Layers. Discuss the importance of these interactions.	5	D24
3	Two most important network-layer functions in a datagram network? Three in a virtual-circuit network?	2	D21-IT
4	What difference does it make to the network layer if the DLL provides connection-oriented vs connectionless service?	2	D21-IT
5	Explain the various design challenges of routing algorithms.	5	D22-CSE
6	State the tunneling concept.	3	D22-IT

Optimality Principle & Shortest Path (Dijkstra) *[HOT] asked 6 times*

#	Question	M	Paper(s)
7	Define routing algorithms in computer networks.	2	D22-ECE
8	Describe / explain the Optimality Principle in routing.	2	N25, D21-ECE
9	Explain Dijkstra's shortest path routing algorithm with an example.	5	D23-CET, D21-ECE
10	Define optimality principle. Illustrate Dijkstra's algorithm on the given graph to find the shortest path from node 1 to node 6.	6	D23-CSE

Flooding *(asked 3 times)*

#	Question	M	Paper(s)
11	Analyze how flooding works and why it can lead to inefficiencies in a network.	5	D24
12	What is flooding? What are its disadvantages? Explain measures to avoid/decrease its impact.	4	D23-CET
13	What is flooding? Explain with an example; mention advantages and disadvantages.	3	D23-CSE

Distance Vector Routing & Count-to-Infinity *[HOT] asked 7 times*

#	Question	M	Paper(s)
14	Explain Distance Vector Routing with an example.	4-5	D23-CET, D22-IT, D21-IT, D21-ECE
15	Examine the distance vector routing algorithm.	5	D22-ECE
16	What is the Count-to-Infinity problem? Where do we get it and what are the solutions?	5	N25
17	What is Count-to-Infinity in DVR? List two approaches to overcome it.	3	D23-CSE

Link State Routing & DV vs LS *[HOT] asked 7 times*

#	Question	M	Paper(s)
18	Explain link state routing (algorithm) with an example / in detail.	4-5	D23-CET, D23-IT, D22-IT
19	Discuss / explain the steps involved in the link state routing algorithm.	5	D22-ECE, D21-ECE
20	State the major difference between distance vector routing and link state routing.	5	N25
21	Differentiate between Distance Vector and Link State routing protocols with examples.	5	M25

OSPF, BGP, Hierarchical Routing *(asked 3 times)*

#	Question	M	Paper(s)
22	Basis of classification for the four types of links defined by OSPF? Why do OSPF messages propagate faster than RIP?	4	D22-IoT
23	What is the use of BGP?	2	J22
24	Distinguish between Intra-AS and Inter-AS routing.	5	D21-IT

Quality of Service & Traffic Shaping *[HOT] asked 9 times*

#	Question	M	Paper(s)
25	Explain Leaky bucket and Token bucket algorithms.	4	D23-CET
26	How does the leaky bucket algorithm improve Quality of Service?	3	D22-IT
27	What is traffic shaping? Explain two methods to shape traffic.	5	M25
28	Define QoS. Elaborate the characteristics of QoS.	5	J22
29	Short notes on data communication characteristics: (i) Reliability (ii) Jitter (iii) Bandwidth.	6	J22
30	Examine the relationship between average data rate and peak data rate; impact on performance and bandwidth allocation.	5	M25
31	How is Resource Reservation Protocol related to Integrated Services? Explain with example.	4	D22-IoT
32	**[Numerical]** Token bucket: new token every 5 microseconds, each token = one 48-byte packet. Maximum sustainable data rate?	5	J23
33	**[Numerical]** Leaky bucket: output 5 gal/min, input burst 100 gal/min for 12 s, then no input for 48 s. Gallons left?	9	J24

IPv4 Protocol: Header, Fragmentation, Checksum *[HOT] asked 7 times*

#	Question	M	Paper(s)
34	Draw the header format of IPv4 and explain the fields in it.	5	D21-IT
35	What are the key fields in an IP datagram?	2	D22-ECE
36	Explain the need for options in IPv4 and list the options with a brief description.	5	J23
37	Explain the procedure for checksum calculation and verification in IPv4.	5	J24
38	What part of an IPv4 packet is covered in the checksum? Why? Are options included?	4	J24
39	Elaborate the concept of packet fragmentation.	4	D22-IT
40	**[Numerical]** 1000-byte IP datagram, MTU 100 bytes, IP header 20 bytes. Number of fragments?	5	D23-IT

IP Addresses, Subnetting & NAT *[HOT] asked 13 times*

#	Question	M	Paper(s)
41	What is Network Address Translation (NAT)?	2	D24
42	What is NAT? How can NAT help in address depletion?	5	J24
43	Describe NAT and its purpose. Give situations where NAT is essential.	5	M25
44	Number of bits in an IPv4 address and in an IPv6 address?	3	D22-IoT
45	Format, bit representation and address space of IPv4 and IPv6.	4	D23-CSE
46	Explain the difference between classful and classless addressing in IPv4.	3	D22-CSE
47	Explain subnetting and supernetting. How do subnet and supernet masks differ from a default mask in classful addressing?	4-5	D22-IoT, J23
48	While IP addresses are tied to specific networks, Ethernet addresses are not. Why?	3	J23

#	Question	M	Paper(s)
49	**[Numerical]** One address of a block is 205.16.37.39/28. Find first address, last address and number of addresses.	5	N25
50	**[Numerical]** C1: 203.197.2.53 / 255.255.128.0; C2: 203.197.75.201 / 255.255.192.0. Same network or different?	5	D22-IT
51	**[Numerical]** Find the class of: 208.34.54.12, 238.34.2.1, 114.34.2.8, 129.14.6.8.	4	J24
52	**[Numerical]** Binary to dotted-decimal: 10000001 00001011 00001011 11101111; 11000001 10000011 00011011 11111111.	4	D23-IT

IP Version 6 *[HOT] asked 9 times*

#	Question	M	Paper(s)
53	How does IPv6 address the problem of IP address exhaustion faced by IPv4?	5	D24
54	Why is IPv6 preferred over IPv4?	2-3	J22, J24
55	Why is IPv4 to IPv6 transition required?	4	D23-IT
56	List three IPv4-to-IPv6 transition strategies. Difference between tunneling and dual stack; when is each used?	4	D22-IoT
57	Explain the IPv6 datagram with a neat diagram.	5	D22-ECE
58	Illustrate the message format and fixed header of IPv6.	5	D22-CSE
59	Distinguish between IPv4 and IPv6.	5	J22
60	Compare the scalability of IPv4 and IPv6 in modern network infrastructure.	5	M25

UNIT IV · Transport Layer

Transport Service & Elements of Transport Protocols *[HOT] asked 17 times*

#	Question	M	Paper(s)
1	What are two essential elements of a transport protocol?	2	D24
2	Explain / summarize the elements of the transport layer.	5	D24, D22-ECE
3	List the functions of the transport layer.	3	D23-IT
4	Discuss the purpose of the transport layer in a computer network.	4	D23-IT
5	What is process-to-process communication? Design issues of the transport layer.	5	J23
6	Explain transport-layer multiplexing and demultiplexing.	3-5	D23-IT, D21-IT, D22-IT
7	Do port addresses need to be unique? Why or why not? Why are port addresses shorter than IP addresses?	3-5	D22-IoT, J23, M25
8	Explain TCP service (response) primitives.	5	D22-ECE, D21-ECE
9	The transport layer creates a connection between source and destination. What are the three events involved?	2	J22
10	Illustrate the concept of crash recovery in the transport layer.	4	D22-IT
11	Describe the QoS primitives of the transport layer.	5	D22-IoT

Berkeley Sockets *[HOT] asked 5 times*

#	Question	M	Paper(s)
12	Explain Berkeley Sockets in detail.	5	D23-IT
13	Discuss the Berkeley Socket API. How do TCP and UDP sockets differ?	5	N25
14	List any two differences between TCP sockets and UDP sockets.	2	N25
15	Explain socket programming with UDP with a neat diagram.	5	D22-ECE
16	Illustrate socket programming with TCP.	5	D21-ECE

UDP *[HOT] asked 11 times*

#	Question	M	Paper(s)
17	Draw the UDP header and explain all the fields.	5	D24, D23-IT
18	Discuss UDP characteristics in detail.	4	D23-IT
19	What does UDP provide that is not provided by IP?	2	D22-ECE, D21-ECE
20	Maximum and minimum size of process data that can be encapsulated in a UDP datagram?	2-3	D22-CSE, M25
21	Are UDP and IP unreliable to the same degree? Justify.	5	M25
22	UDP is message-oriented, TCP is byte-oriented. If an application must protect message boundaries, which should it use?	4-5	D22-IoT, J23
23	When would a client prefer RPC over UDP, and when RPC over TCP?	4	J23

TCP vs UDP *[HOT] asked 9 times*

#	Question	M	Paper(s)
24	Compare and contrast TCP and UDP protocols.	2-5	D24, D23-CET, J22
25	Differentiate connectionless and connection-oriented protocols. Which is more efficient and why?	3	J24
26	Compare the TCP header and the UDP header. List fields in TCP missing from UDP and give reasons.	4-5	D23-CSE, D22-CSE, D21-IT
27	Why do HTTP, FTP, SMTP and POP3 run on top of TCP rather than UDP?	2	D21-IT
28	For a fastest-possible remote transaction, would you use UDP or TCP? Why?	2	D21-IT

TCP Protocol & TCP Segment Header *[HOT] asked 9 times*

#	Question	M	Paper(s)
29	Draw / sketch the TCP segment header and explain each field.	4-5	N25, D23-CET, D22-IT, D22-ECE, J22
30	Discuss the role of TCP in providing a robust and stable communication framework.	5	D23-IT
31	Explain the features of TCP.	4	J24
32	IP fragmentation is invisible to TCP. Does TCP still worry about out-of-order data? Explain.	4	D22-CSE
33	Datagram carrying a TCP segment for 130.14.16.17/16 arrives (corrupted port) at 130.14.16.19/16. How does the receiving TCP react?	9	J24

TCP Connection Establishment & Release *[HOT] asked 15 times*

#	Question	M	Paper(s)
34	Explain connection management in the transport layer / how TCP connection management is implemented.	4-5	D24, D23-IT
35	Discuss connection establishment and release in TCP with a suitable diagram.	5	N25, D23-CET
36	Three-way handshake in TCP: how connection establishment and release take place.	5	D23-CSE, J24
37	How does the three-way handshake ensure reliability of TCP?	5	D22-IT
38	Explain various steps involved in releasing a TCP connection.	5	D22-IT
39	Explain TCP connection establishment in the normal case and call-collision case.	5	J24
40	Explain the two-way and three-way handshake mechanisms.	5	D22-ECE
41	If a two-way handshake were used instead of three-way, are deadlocks possible? Example or show none exist.	4-5	J23, D22-CSE
42	What is four-way handshaking? In which protocol is it used?	3	D23-CSE
43	Draw the TCP entity state diagram and explain.	5	D21-ECE
44	Generalized n-army problem (any two blue armies agreeing suffices). Does a protocol exist for blue to win?	5	D22-IoT

TCP Sliding Window, Flow Control & Timers *[HOT] asked 5 times*

#	Question	M	Paper(s)
45	Explain sliding window / window management in TCP.	4	D22-IT, J24
46	Briefly describe the credit scheme used by TCP for flow control.	2	D21-ECE
47	Explain TCP Timer Management in detail.	4	D23-IT
48	List and briefly explain various timers used by TCP.	4	J22

Congestion Control *[HOT] asked 12 times*

#	Question	M	Paper(s)
49	What is congestion? Illustrate how to control congestion in a network.	5	N25

#	Question	M	Paper(s)
50	How does congestion occur in a network?	3	D23-IT
51	Analyze the general principles of congestion control.	4	D23-CET
52	Explain / demonstrate the approaches to congestion control.	5	D22-IoT, D21-IT
53	Explain the mechanism in the transport layer for controlling congestion.	5	D22-IT
54	Compare flow control, error control and congestion control.	4	D22-IT
55	Difference between open-loop and closed-loop congestion control. Policies that prevent congestion?	4	D22-CSE
56	How can TCP be used to deal with network or internet congestion?	2 / 5	D22-ECE, D21-ECE
57	Advantages and disadvantages of congestion control at the transport layer rather than the network layer.	5	D21-ECE
58	**[Numerical]** TCP AIMD: slow start window 2 MSS, threshold 8 MSS, timeout in 5th transmission. Congestion window at end of 10th transmission?	5	D22-IT

UNIT V · Application Layer

DNS: Name Space, Resource Records, Name Servers *[HOT] asked 21 times*

#	Question	M	Paper(s)
1	What is a domain resource record in DNS? / Short note on Domain Resource Records.	2 / 4	D24, D23-IT
2	Examine the key components of DNS and the role each plays.	5	D24
3	What are the three main components of DNS?	2	J22
4	Describe the role of DNS with reference to its components and working.	5	J22
5	Explain DNS / discuss Domain Name System in detail.	3 / 5	D23-CET, D22-IT
6	Give the details of / describe the services provided by DNS.	5	D22-ECE, D21-ECE, D21-IT
7	Why do we need a DNS system when we can directly use an IP address?	5	M25
8	Advantage of a hierarchical name space over a flat name space for the Internet?	3	D22-IoT
9	What is FQDN and PQDN? Give an example of each.	2	N25
10	Compare DNS iterative and recursive resolution with examples.	3 / 5	N25, J23
11	Two main categories of DNS messages (with example).	5	D22-IoT, J23, D22-CSE
12	For voyager.fhda.edu, how many labels and how many levels of hierarchy?	(part)	J23
13	Why was there a need for DDNS?	2	M25, D22-IoT

Electronic Mail: Architecture, User Agent, Transfer, Delivery *[HOT] asked 8 times*

#	Question	M	Paper(s)
14	What are the major components in an e-mail system? Explain.	5	D24
15	Illustrate the architecture of e-mail.	5	N25
16	Explain the working of electronic mail / e-mail in detail.	5-7	D23-CET, D21-ECE
17	Architectural layout of e-mail; functionality of User Agent (UA) and Message Transfer Agent (MTA).	4	D23-CSE
18	In electronic mail, explain the tasks of a user agent.	3	D22-CSE
19	How does the user get e-mail from the message transfer agent?	4	D22-IT
20	Why is connection establishment for mail transfer needed if TCP already established a connection?	4	D22-CSE

SMTP, POP3, IMAP, MIME *[HOT] asked 6 times*

#	Question	M	Paper(s)
21	Describe how SMTP is used in e-mail applications.	5	D23-IT
22	Illustrate the basic operation of SMTP and compare it with HTTP.	5	D21-ECE
23	Describe the addressing system used by SMTP. Why do we need POP3 or IMAP4?	3-4	J23, J24
24	Differentiate IMAP and POP with a real-time example.	3	D23-CSE
25	Write short notes on IMAP and MIME with an example.	5	N25

FTP, TELNET, SNMP *[HOT] asked 11 times*

#	Question	M	Paper(s)
26	Write a brief note on FTP.	5	J22
27	What is the need of FTP? Compare it with HTTP.	5	D22-ECE
28	Compare HTTP and FTP by functionality and data transfer; scenarios for each.	5	M25

#	Question	M	Paper(s)
29	List out the FTP commands.	2	D22-ECE
30	Three FTP transmission modes? How does storing a file differ from retrieving?	4	D22-IoT
31	Differentiate FTP and SFTP.	3	D23-CSE
32	Evaluate limitations on anonymous FTP access; risks and how it may be exploited.	5	M25
33	With a neat diagram explain the architecture of SNMP.	5	J22
34	SNMP: working principle and related issues.	6	D23-CSE
35	Define fault management and its purpose; explain its two sub-functions.	4	J23
36	List the Application Layer protocols.	3	D23-IT

APPENDIX · Asked in papers, outside your listed syllabus

Units I–V above contain only questions on topics named in your syllabus. Everything here is outside it: the first table is close to a unit (Ethernet, ICMP, multicast routing, etc.) and is worth a skim; the rest is clearly out. Note: WWW, streaming and ICMP questions did appear in recent R22 papers.

Borderline: related to a unit but topic not named in syllabus *[HOT] asked 24 times*

#	Question	M	Paper(s)
1	Compare the telephone network and the Internet. What are the similarities and differences? [Unit I]	4	J23
2	Sketch the Manchester encoding for the bit stream 0001110101. [Unit I]	2	J22
3	Six stations A-F use MACA. Can two transmissions take place simultaneously? Explain. [Unit Ii]	5	D22-CSE
4	Demonstrate collision avoidance using RTS and CTS frames in 802.11 MAC with a neat diagram. [Unit Ii]	5	D21-ECE
5	Explain the Ethernet. [Unit Ii]	5	N25
6	Illustrate the Ethernet frame structure and explain Ethernet technologies. [Unit Ii]	5	D22-ECE
7	Describe Ethernet frame format with a neat diagram. [Unit Ii]	5	D21-IT
8	Draw and explain IEEE 802.3 frame format. [Unit Ii]	5	J22
9	What is the broadcast address for Ethernet? [Unit Ii]	2	M25
10	If an Ethernet destination address is 07:01:02:03:04:05, is it unicast, multicast or broadcast? [Unit Ii]	3	J23
11	Describe the structure and function of a router in a network. [Unit Iii]	5	M25
12	Distinguish between broadcast and multicast routing protocols. [Unit Iii]	5	D21-ECE
13	Describe various broadcast routing techniques. [Unit Iii]	5	D21-IT
14	Explain multicast routing algorithm. [Unit Iii]	5	D22-ECE
15	List some practical applications of multicasting. [Unit Iii]	2	D21-ECE
16	Compute a multicast spanning tree for router C for a group with members at routers A, B, C, D, E, F, I, K (given graph). [Unit Iii]	4	D22-CSE
17	Explain unicast and multicast forwarding protocols for routing. [Unit Iii]	5	J22
18	Write a short note on ICMP / explain the working of ICMP. [Unit Iii]	4-5	N25, D22-CSE
19	Discuss the role of ICMP. List different ICMP message types. [Unit Iii]	5	D23-CSE
20	What are the different types of messages used by ICMP? [Unit Iii]	2	D21-IT
21	Differentiate HTTP with SMTP. [Unit V]	2	D22-ECE, D21-ECE
22	How is HTTP similar to SMTP? What is a proxy server and how is it related to HTTP? [Unit V]	5	D22-IoT

WWW, HTTP & Web *[HOT] asked 10 times*

#	Question	M	Paper(s)
23	Write short notes on World Wide Web / explain its working and architecture.	3-7	D24, D23-CET, D23-IT, D22-IT, D23-CSE
24	Explain the working of HTTP.	2	D23-CET
25	List out the HTTP commands.	2	D21-ECE
26	Compare HTTP with persistent and non-persistent connection.	2	J22
27	Explain the process of loading a given web page (IP unknown, HTTP 1.1).	5	D23-IT
28	Explain the three common components of a browser.	5	J23

Multimedia & CDN *[HOT] asked 5 times*

#	Question	M	Paper(s)
29	Explain about streaming audio.	5	D24
30	Describe the concept of streaming stored media.	5	D22-IT
31	Differentiate streaming stored media and streaming live media with use cases.	5	N25
32	Does voice over IP have the same problems with firewalls that streaming audio does?	4	D22-IoT
33	Outline the concept of Content Delivery Networks.	4	D22-IT

Security *[HOT] asked 8 times*

#	Question	M	Paper(s)
34	What is a firewall and why is it used? / Explain packet-filter firewall.	2 / 5	D23-CET, M25
35	Public key cryptography: which statement is TRUE (sender/receiver keys)?	2	D21-IT
36	Short notes on VPN.	2	D21-IT
37	Hash functions and Message Authentication Code; RSA with example; IPsec; two symmetric key techniques.	5 each	D21-IT
38	Short notes on PGP, SSH. / How does PGP create security parameters?	4-5	J22, D22-CSE
39	Symmetric keys needed for 10 people communicating with another group of 10.	5	D22-CSE

Other Link / Network Layer Topics *[HOT] asked 14 times*

#	Question	M	Paper(s)
40	HDLC frame format and control field (I, S, U frames).	4-5	J22, D23-CSE
41	Compare HDLC with PPP; which is byte-oriented, which bit-oriented?	4	D22-CSE
42	Characteristics of frame relay.	2	J22
43	Bridges vs Routers / relationship between switch and bridge; Bluetooth slot hopping times.	3-5	D21-ECE, D22-CSE, J23
44	Principal components of 802.11 wireless LAN architecture.	5	D22-ECE
45	How VLANs help reduce network traffic.	5	M25
46	Link layer addressing and ARP; RARP and ARP implementation.	4-5	D21-IT, J23
47	DHCP; IGMP message format and operation.	5	J22
48	SCTP connection establishment; association; advantages over TCP and UDP.	3-6	D23-CSE
49	Short notes on Data Center Networking.	2	D21-IT